Bulk RNA-seq Analysis

• Run raw-read QC and interpret key metrics.

• Perform RNA-seq mapping or pseudoalignment.

• Generate gene-level count matrices.

• Perform differential expression (DE) analysis.

• Interpret PCA, MA plot, volcano plot, and enriched pathways (optional).

• Produce a reproducible report (R Markdown / Quarto).

Dataset description

You're using the **HBR/UHR/ERCC RNA-seq dataset**

- **Biological sources**
  - **HBR** = Human Brain Reference RNA
  - **UHR** = Universal Human Reference RNA
- **Spike-ins**
  - **ERCC** synthetic RNA controls are included
  - HBR libraries use one ERCC mix, UHR libraries use the other (commonly Mix2 vs Mix1 in file names)
- **Replicates**
  - 3 replicates per group:
    - HBR_Rep1, HBR_Rep2, HBR_Rep3
    - UHR_Rep1, UHR_Rep2, UHR_Rep3
- **Read type**
  - **Paired-end** FASTQ files (`read1` and `read2` for each replicate)

What the file names mean

Example:
`UHR_Rep3_ERCC-Mix1_Build37-ErccTranscripts-chr22.read2.fastq.gz`

- `UHR` → sample group (UHR vs HBR)
- `Rep3` → biological replicate number
- `ERCC-Mix1` → spike-in mix used
- `Build37-ErccTranscripts-chr22` → tutorial reference subset context

- `read2` → mate 2 of paired-end sequencing
- `.fastq.gz` → gzipped FASTQ format

Total sample structure in your folder

- 12 FASTQ files total:
    - 6 samples × 2 reads each (R1/R2)
- Balanced design:
    - HBR: 3 replicates
    - UHR: 3 replicates

Part 1 — Raw read QC (FastQC + MultiQC)

Tasks

1. Run FastQC on each FASTQ.
2. Aggregate using MultiQC.
3. Fill a QC table:
    - total reads
    - per-base quality
    - adapter content
    - overrepresented sequences
    - GC bias flags

Questions

- Which samples show adapter contamination?
- Would you trim? Why?
- Which QC warning is acceptable for RNA-seq and why?

Minimum machine requirements

For a lightweight class dataset (4–8 samples, small FASTQ files):

- **RAM:** 8 GB minimum (16 GB recommended)
- **Disk free:** at least 20–30 GB
- **CPU:** 4 cores recommended
- **OS:**
    - macOS (Intel/Apple Silicon) ✅
    - Linux ✅
    - Windows ✅ (use WSL2 Ubuntu)

1. Download Miniconda (official)

Go to the official Miniconda page and download the installer for your OS/CPU.

https://www.anaconda.com/docs/getting-started/miniconda/main

Install by OS

macOS (Terminal)

1. Download the right installer (`arm64` for Apple Silicon, `x86_64` for Intel).
2. In Terminal:

```
bash Miniconda3-latest-MacOSX-<arch>.sh
```

(Replace <arch> with `arm64` or `x86_64`.)

3. Accept license, choose install path, allow `conda init` when prompted.
4. Close and reopen Terminal.

Linux (Terminal)

1. Download Linux installer (`x86_64` or `aarch64`).
2. Run:

```
bash Miniconda3-latest-Linux-<arch>.sh
```

3. Accept prompts and allow initialization.
4. Close and reopen Terminal.

Windows

1. Download Windows Miniconda installer (`.exe`).
2. Double-click installer and follow wizard defaults.
3. Open **Miniconda Prompt** after install.

Verify installation

Run:

```
conda --version
conda list
```

In Terminal:

```
# Check chip type first
uname -m
```

- If output is `arm64` (Apple Silicon), use arm installer.
- If output is `x86_64` (Intel), use Intel installer.

Then run:

```
# Apple Silicon (M1/M2/M3)
curl -LO https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-
arm64.sh
bash Miniconda3-latest-MacOSX-arm64.sh

# OR Intel Mac
curl -LO https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-
x86_64.sh
bash Miniconda3-latest-MacOSX-x86_64.sh
```

After install:

```
conda init zsh
exec zsh
conda –version
```

If Miniconda is installed correctly.

Run these commands **exactly** in your terminal:

conda install -n base -c conda-forge mamba -y

mamba install -y -c conda-forge -c bioconda \

fastqc multiqc fastp salmon subread samtools \

r-base r-essentials r-tidyverse r-data.table \

bioconductor-deseq2 bioconductor-tximport \

bioconductor-apeglm bioconductor-pheatmap

Now do these **exact steps to** get your RNA-seq environment working.

Accept Conda Terms once

Run:

```
conda tos accept --override-channels --channel
https://repo.anaconda.com/pkgs/main
conda tos accept --override-channels --channel
https://repo.anaconda.com/pkgs/r
```

If successful, no error message.

Set channels (recommended for bioinformatics)

```
conda config --add channels conda-forge
conda config --add channels bioconda
conda config --add channels defaults
conda config --set channel_priority strict
```

Check:

```
conda config --show channels
```

You should see channels including `conda-forge`, `bioconda`, `defaults`.

Create clean environment for class

```
conda create -n rnaseq_lab python=3.11 -y
```

Then activate it:

```
conda activate rnaseq_lab
```

You should now see `(rnaseq_lab)` at the start of your prompt.

Install required tools in that environment

```
conda install -y \
  fastqc multiqc fastp salmon subread samtools \
  r-base r-essentials r-tidyverse r-data.table \
  bioconductor-deseq2 bioconductor-tximport \
  bioconductor-apeglm bioconductor-pheatmap
```

If solving is slow, do this first in **base**:

```
conda activate base
conda install -n base -c conda-forge mamba -y
conda activate rnaseq_lab
mamba install -y \
  fastqc multiqc fastp salmon subread samtools \
  r-base r-essentials r-tidyverse r-data.table \
  bioconductor-deseq2 bioconductor-tximport \
  bioconductor-apeglm bioconductor-pheatmap
```

Verify tools are installed

With `rnaseq_lab` active, run:

```
which fastqc
which multiqc
which fastp
which salmon
which samtools
which featureCounts
R --version
python --version
```

All commands should return paths/versions (not “command not found”).

Create your project folder

```
mkdir -p ~/rnaseq_lab/{data,ref,qc,trimmed,quant,counts,results,logs,scripts}
cd ~/rnaseq_lab
```

Make sure you are in the right env

```
conda activate rnaseq_lab
which python
conda info --envs
```

You should see `rnaseq_lab` active and python path inside `.../envs/rnaseq_lab/...`.

Install mamba (faster)

```
conda activate base
conda install -n base -c conda-forge mamba -y
```

Re-activate your lab env

```
conda activate rnaseq_lab
```

Install missing tools (using mamba)

Run this exact command:

```
mamba install -y -c conda-forge -c bioconda \
  fastqc multiqc fastp salmon subread samtools
```

Then install R/Bioconductor packages (if needed):

```
mamba install -y -c conda-forge -c bioconda \
  r-base r-essentials r-tidyverse r-data.table \
  bioconductor-deseq2 bioconductor-tximport \
  bioconductor-apeglm bioconductor-pheatmap
```

Verify again

```
which fastqc
which multiqc
which fastp
which salmon
which samtools
which featureCounts
fastqc --version
multiqc --version
fastp --version
salmon --version
samtools --version
featureCounts -v
R --version
python --version
```

If commands still "not found"

Run:

```
echo $PATH
conda run -n rnaseq_lab fastqc --version
conda run -n rnaseq_lab salmon --version
```

- If `conda run ...` works but plain command doesn’t, shell init is the issue. Fix with:

```
conda init zsh
exec zsh
conda activate rnaseq_lab
```

Last-resort clean rebuild (only if needed)

If installs keep failing, rebuild environment once:

```
conda deactivate
conda env remove -n rnaseq_lab -y
mamba create -n rnaseq_lab -y -c conda-forge -c bioconda \
  python=3.11 fastqc multiqc fastp salmon subread samtools \
  r-base r-essentials r-tidyverse r-data.table \
  bioconductor-deseq2 bioconductor-tximport \
  bioconductor-apeglm bioconductor-pheatmap
conda activate rnaseq_lab
```

Also, `pheatmap` is a CRAN package (not Bioconductor), so this is expected.

Run this exact command now:

```
mamba install -y -c conda-forge -c bioconda \
  r-base r-essentials r-tidyverse r-data.table r-pheatmap \
  bioconductor-deseq2 bioconductor-tximport bioconductor-apeglm
```

Then verify everything:

```
which fastqc
which multiqc
which fastp
which salmon
which samtools
which featureCounts
R --version
python --version
```

And test R libraries:

```
R -q -e "library(DESeq2); library(tximport); library(apeglm); library
```

Run these **exactly**:

```
# 1) make sure you are in the right env
conda activate rnaseq_lab

# 2) install Java explicitly in this env
mamba install -y -c conda-forge openjdk=21
```

Now test:

```
which java
java -version
```

If `java -version` works, test FastQC:

```
fastqc --version
```

If `java` is still not found, force-link JAVA_HOME to the env's JDK:

```
conda activate rnaseq_lab
export JAVA_HOME="$CONDA_PREFIX"
export PATH="$JAVA_HOME/bin:$PATH"

which java
java -version
fastqc --version
```

If that works, persist it for every activation:

```
mkdir -p "$CONDA_PREFIX/etc/conda/activate.d"

cat > "$CONDA_PREFIX/etc/conda/activate.d/java.sh" << 'EOF'

export JAVA_HOME="$CONDA_PREFIX"
export PATH="$JAVA_HOME/bin:$PATH"
EOF
```

Then reload env:

```
conda deactivate
conda activate rnaseq_lab
java -version
fastqc --version
```

On macOS, Java can be installed in the env but not automatically visible to shell lookup. Setting `JAVA_HOME` + PATH inside the conda env fixes that reliably.

After Java is fixed, run this (last R package step with correct name):

```
mamba install -y -c conda-forge -c bioconda \
  r-tidyverse r-data.table r-pheatmap \
  bioconductor-deseq2 bioconductor-tximport bioconductor-apeglm
```

Just one Mac note: `wget` may not be installed by default. If it fails, use `curl`.

Use this sequence:

```
# 1) set folder
export RNA_DATA_DIR=~/rnaseq_qc_lab/data
mkdir -p "$RNA_DATA_DIR"
cd "$RNA_DATA_DIR"

# 2) download (choose ONE)
# wget version:
wget http://genomedata.org/rnaseq-tutorial/HBR_UHR_ERCC_ds_5pc.tar

# OR curl version (works on mac by default):
curl -L -o HBR_UHR_ERCC_ds_5pc.tar http://genomedata.org/rnaseq-
tutorial/HBR_UHR_ERCC_ds_5pc.tar

# 3) extract
tar -xvf HBR_UHR_ERCC_ds_5pc.tar

# 4) check files
find "$RNA_DATA_DIR" -type f | head -20
find "$RNA_DATA_DIR" -type f \( -name "*.fastq.gz" -o -name "*.fq.gz" -o -
name "*.fastq" -o -name "*.fq" \)
```

Then run QC from lab root:

```
cd ~/rnaseq_qc_lab
mkdir -p qc/fastqc_raw qc/multiqc

# if files are gzipped
fastqc data/*.fastq.gz -o qc/fastqc_raw -t 4

# if files are NOT gzipped, use:
# fastqc data/*.fastq -o qc/fastqc_raw -t 4

multiqc qc/fastqc_raw -o qc/multiqc
open qc/multiqc/multiqc_report.html
```

If your FASTQs are inside a subfolder after extraction, run this instead:

```
cd ~/rnaseq_qc_lab
find data -type f \( -name "*.fastq.gz" -o -name "*.fq.gz" -o -name "*.fastq"
-o -name "*.fq" \) -print0 | xargs -0 fastqc -o qc/fastqc_raw -t 4
multiqc qc/fastqc_raw -o qc/multiqc
open qc/multiqc/multiqc_report.html
```

If you paste the output of:

```
find ~/rnaseq_qc_lab/data -type f | head -30
```

dataset is in place and looks correct (6 paired-end samples: HBR x3, UHR x3).
Now run QC with this exact command pattern.

```
conda activate rnaseq_lab
cd ~/rnaseq_qc_lab
mkdir -p qc/fastqc_raw qc/multiqc

# FastQC on all gzipped FASTQ files in data/
fastqc data/*.fastq.gz -o qc/fastqc_raw -t 4

# Aggregate to one HTML report
multiqc qc/fastqc_raw -o qc/multiqc

# Open report
open qc/multiqc/multiqc_report.html
```

to avoid analyzing the `.tar` file by mistake, this is also clean:

```
find data -maxdepth 1 -type f -name "*.fastq.gz" -print0 | xargs -0 fastqc -o
qc/fastqc_raw -t 4
multiqc qc/fastqc_raw -o qc/multiqc
open qc/multiqc/multiqc_report.html
```

in MultiQC:

1. **Per base sequence quality** (are tails dropping?)
2. **Per sequence quality scores**
3. **Adapter content**
4. **Sequence duplication levels**
5. **Overrepresented sequences**
6. **Per base N content**
7. **GC distribution**

Short interpretation prompt

- What are the top 2–3 QC concerns?
- Are issues mild/moderate/severe?
- Would you proceed directly, or trim first? Why?
- Which samples look worst/best?

For each sample, summarize:

1) Per base sequence quality

- Are most positions in green (good)?
- If quality drops, at what read position range?

2) Adapter content

- Is adapter contamination present?
- Is it low, moderate, or high?
- Is trimming recommended?

3) Sequence duplication levels

- Low/moderate/high duplication?
- Is this plausible for RNA-seq (highly expressed transcripts) or likely technical artifact?

4) Overrepresented sequences

- Are there flagged sequences?
- Likely adapter/rRNA/other contamination?

5) GC content

- Does GC look roughly unimodal/expected?
- Any unusual shift across samples?

MultiQC is the **summary dashboard**.
You can still view each sample's **original FastQC report** (`.html`) directly.

Open FastQC reports from Finder

1. Go to:
   `~/rnaseq_qc_lab/qc/fastqc_raw`
2. You'll see files like:
   - `..._fastqc.html`
   - `..._fastqc.zip`
3. Double-click any `*_fastqc.html` to open that sample's full FastQC report.
4.

Open from terminal (Mac)

5. `cd ~/rnaseq_qc_lab/qc/fastqc_raw`
6. `ls -1 *_fastqc.html`
7. `open *_fastqc.html`
8. If that opens too many tabs, open one sample:
9. `open HBR_Rep1_ERCC-Mix2_Build37-ErccTranscripts-chr22.read1_fastqc.html`

From MultiQC links

In MultiQC:

- Go to a FastQC section/table (or “General Stats”).
- Click a sample name/module entry — MultiQC often links to the underlying FastQC file.
- If popups are blocked, allow popups for local files/browser.